

File Edit Selection View Go Run Terminal Helppython_test

EXPLORER

PYTHON_TEST

__pycache__

.vscode

settings.json

New folder

annotated_image.jpg

black.jpg

cat.png

clock.png

create_test_image_day3.py

create_test_image.py

crop.py

DAY-01 Program Outputs.pdf

DAY-02 Program outputs.pdf

DAY-03 Program Outputs.pdf

DAY-04 Program Outputs.pdf

DAY-05 Program Outputs.pdf

day5_transformations.py

day6_arithmetic_bitwise.py

drawing_basics.py

drawing.py

exercise1.day3.py

exercise1.house.py

exercise1.py

exercise1.thumbnail.py

exercise2.day3.py

exercise2.panorama.py

exercise2.py

exercise2.target.py

exercise3.animation.py

exercise3.chessboard.py

exercise3.day3.py

exercise3.py

exercise4.annotations.py

OUTLINE

TIMELINE

Welcome

day6_arithmetic_bitwise.py

day6_arithmetic_bitwise.py > ...

1 # day6_arithmetic_bitwise.py

2 # Complete guide to image arithmetic and bitwise operations

3 from __future__ import print_function

4 import numpy as np

5 import argparse

6 import cv2

7 print("=" * 70)

8 print("DAY 6: IMAGE ARITHMETIC & BITWISE OPERATIONS")

9 print("=" * 70)

10 # -----

11 # SECTION 1: UNDERSTANDING OPENCV VS NUMPY ARITHMETIC

12 # -----

13 print("\n[Section 1] OpenCV vs NumPy Arithmetic")

14 print("-" * 50)

15 # Create simple test values

16 value1 = np.uint8([200])

17 value2 = np.uint8([100])

18 print(f"Test values: {value1[0]} + {value2[0]} = ?")

19 print(f"Test values: {value1[0]} - {value2[0]} = ?")

20 print("\n--- OpenCV (Clipping) ---")

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

OpenCV: 10 - 20 = [-10.]

NumPy: 10 - 20 = 246

OpenCV: 255 + 1 = [256.]

NumPy: 255 + 1 = 0

OpenCV: 255 - 1 = [254.]

NumPy: 255 - 1 = 254

OpenCV: 0 + 1 = [1.]

NumPy: 0 + 1 = 1

OpenCV: 0 - 1 = [-1.]

NumPy: 0 - 1 = 255

[Section 2] Loading images for arithmetic operations...

Images have different sizes: (228, 350, 3) vs (486, 474, 3)

Resizing image2 to match image1...

Both images now: 350 x 228 pixels

Image 2



Image 1



python

python



File

Edit

Selection

View

Go

Run

Terminal

Help

python_test

EXPLORER

PYTHON TEST

dissoive.mp4

drawing_basics.py

drawing.py

exercise1.brightness.c...

exercise1.day3.py

exercise1.house.py

exercise1.py

exercise1.thumbnail.py

exercise2.day3.py

exercise2.dissolve.py

exercise2.panorama.py

exercise2.py

exercise2.target.py

exercise3.animation.py

exercise3.chessboard.py

exercise3.day3.py

exercise3.py

exercise3.toolmask.py

exercise4.annotations....

exercise4.day3.py

exercise4.smart.py

exercise5.alignment.py

exercise5.clock.py

flipping.py

getting_and_setting.py

gradient.png

house.png

image.py

imutils.py

load_display_save.py

masked_result.png

my_masterpiece.png

output.bmp

OUTLINE

TIMELINE

Welcome

exercise1.brightness.contrast.py

exercise2.dissolve.py

exercise3.toolmask.py

exercise3.toolmask.py > ...

1 # exercise_3_mask_tool.py

2

3 import cv2

4 import numpy as np

5

6 # Load image

7 image = cv2.imread("test_image.png")

8

9 if image is None:

10 print("Error: Could not load test_image.png")

11 exit()

12

13 # Create empty mask

14 mask = np.zeros(image.shape[:2], dtype="uint8")

15

16 drawing = False

17 ix, iy = -1, -1

18

19

20 def draw_rectangle(event, x, y, flags, param):

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise3.toolmask.py

Draw a rectangle on the image to create a mask

Press 's' to save mask

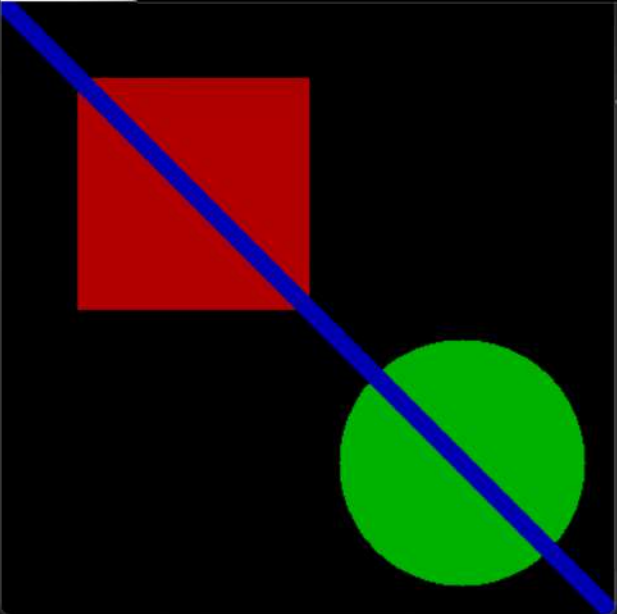
Press 'r' to reset mask

Press ESC to exit

Saved mask and result!

Mask reset

Masked Result



... | [] x

powersh... ⚠

powersh... ⚠

powersh... ⚠

powersh... ⚠

powersh... ⚠

powersh... ⚠

powershell

powershell

powershell

powershell

powershell

powershell

python

Ln 89, Col 24

Spaces: 4

UTF-8

CRLF

{ }

Python

3.14.5

Python 3.14 (64-bit)

File Edit Selection View Go Run Terminal Help python_test

Update

EXPLORER

- PYTHON_TEST
 - dissolve.mp4
 - drawing_basics.py
 - drawing.py
 - exercise1.brighthness.c...
 - exercise1.day3.py
 - exercise1.house.py
 - exercise1.py
 - exercise1.thumbnail.py
 - exercise2.day3.py
 - exercise2.dissolve.py
 - exercise2.panorama.py
 - exercise2.py
 - exercise2.target.py
 - exercise3.animation.py
 - exercise3.chessboard.py
 - exercise3.day3.py
 - exercise3.py
 - exercise3.toolmask.py
 - exercise4.annotations...
 - exercise4.day3.py
 - exercise4.smart.py
 - exercise5.alignment.py
 - exercise5.clock.py
 - flipping.py
 - getting_and_setting.py
 - gradient.png
 - house.png
 - image.py
 - imutils.py
 - load_display_save.py
 - masked_result.png
 - my_masterpiece.png
 - output.htm

exercise1.brighthness.contrast.py > ...

```
91 | bright = max(-255, bright - 10)
92 |
93 | elif key == ord('c'):
94 |     cont = min(127, cont + 10)
95 |
96 | elif key == ord('C'):
97 |     cont = max(-127, cont - 10)
98 |
99 | elif key == ord('r'):
100 |     bright, cont = 0, 0
101 |
102 | elif key == 27: # ESC
103 |     break
104 |
105 | cv2.destroyAllWindows()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

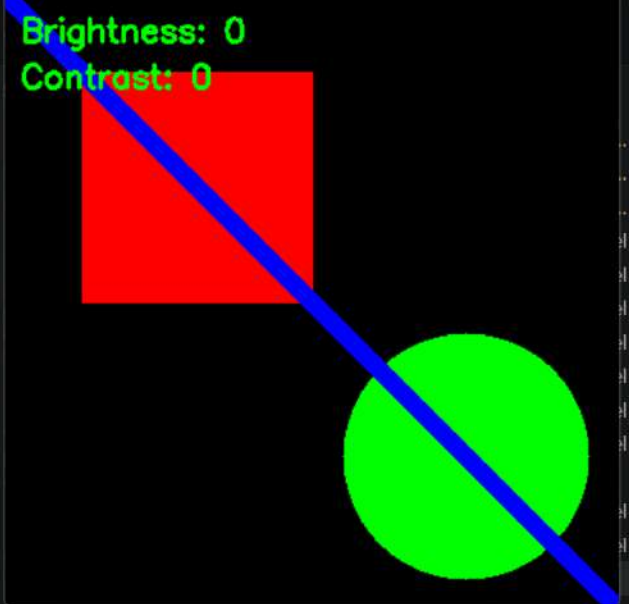
PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise1.brighthness.contrast.py

Press keys to adjust:

- b = increase brightness
- B = decrease brightness
- c = increase contrast
- C = decrease contrast
- r = reset
- ESC = exit

Adjust Brightness/Contrast

Brightness: 0
Contrast: 0



File Edit Selection View Go Run Terminal Help python_test

EXPLORER

PYTHON_TEST

- custom_mask.png
- DAY-01 Program Outp...
- DAY-02 Program out...
- DAY-03 Program Outp...
- DAY-04 Program Outp...
- DAY-05 Program Outp...
- day5_transformations....
- day6_arithmetic_bitwis...
- dissolve.mp4
- drawing_basics.py
- drawing.py
- exercise1.brightness.c...
- exercise1.day3.py
- exercise1.house.py
- exercise1.py
- exercise1.thumbnail.py
- exercise2.day3.py
- exercise2.dissolve.py
- exercise2.panorama.py
- exercise2.py
- exercise2.target.py
- exercise3.animation.py
- exercise3.chessboard.py
- exercise3.day3.py
- exercise3.py
- exercise3.toolmask.py
- exercise4.annotations....
- exercise4.day3.py
- exercise4.smart.py
- exercise4.splash.py**
- exercise5.alignment.py
- exercise5.clock.py
- flipping.py

WELCOME

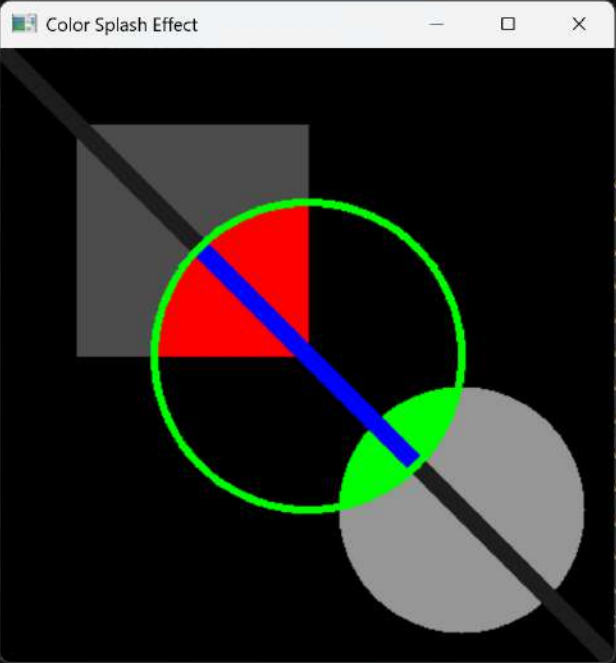
exercise4.splash.py

```
43 image,
44 image,
45 mask=mask
46 )
47
48 # Make outside grayscale
49 outside = cv2.bitwise_and(
50     gray_color,
51     gray_color,
52     mask=mask_inv
53 )
54
55 # Combine both regions
56 splash = cv2.add(inside, outside)
57
58 # Draw circle boundary
59 cv2.circle(
60     splash,
61     center,
62     radius,
63     (0, 255, 0))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise4.splash.py

Color Splash Effect



Ln 77, Col 24 Spaces: 4 UTF-8 CRLF {} Python 3.14.5 Python 3.14 (64-bit)

File Edit Selection View Go Run Terminal Help python_test

EXPLORER

- PYTHON_TEST
 - exercise2.day3.py
 - exercise2.dissolve.py
 - exercise2.panorama.py
 - exercise2.py
 - exercise2.target.py
 - exercise3.animation.py
 - exercise3.chessboard.py
 - exercise3.day3.py
 - exercise3.py
 - exercise3.toolmask.py
 - exercise4.annotations....
 - exercise4.day3.py
 - exercise4.smart.py
 - exercise4.splash.py
 - exercise5.alignment.py
 - exercise5.clock.py
 - exercise5.mask.py
 - flipping.py
 - getting_and_setting.py
 - gradient.png
 - house.png
 - image.py
 - imutils.py
 - load_display_save.py
 - masked_result.png
 - my_masterpiece.png
 - output.bmp
 - output.jpg
 - output.tiff
 - panorama_prep.png
 - pixel_basics.py
 - red.jpg
 - resize.py

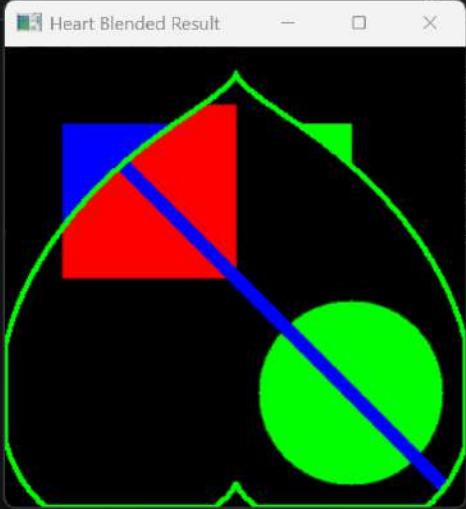
exercise5.mask.py

```
14 image2,  
15 image2,  
16 mask=heart_inv  
17 )  
18  
19 # Combine images  
20 final = cv2.add(result1, result2)  
21  
22 # Draw heart outline  
23 contours, _ = cv2.findContours(  
24     heart,  
25     cv2.RETR_EXTERNAL,  
26     cv2.CHAIN_APPROX_SIMPLE  
27 )  
28  
29 cv2.drawContours(  
30     final,  
31     contours,  
32     -1,  
33     (0, 255, 0),  
34     2
```

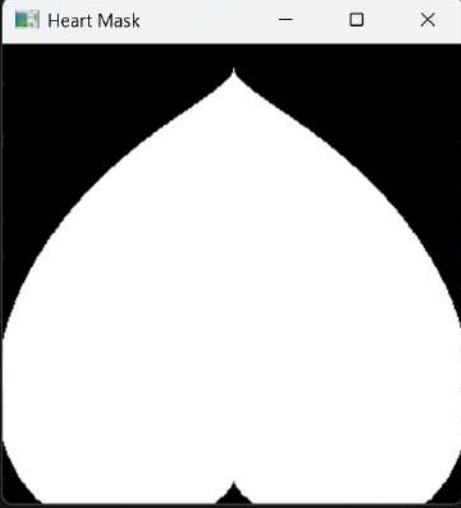
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise5.mask.py

Heart Blended Result



Heart Mask



powershell powershell powershell powershell powershell powershell powershell powershell powershell powershell powershell python